Regularization Structure Deforms the Precision-Recall-FNR Pareto Surface: Evidence from Threshold-Sweep Analysis on the Breast Cancer Benchmark

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Abstract

We ask whether the choice of regularization penalty—none, L1-only, or elastic-net (L1+L2)—reshapes the Pareto frontier in three-dimensional precision-recall-false-negative-rate (FNR) space for malignant-class detection on the `sklearn` breast cancer benchmark. We frame "non-trivial reshaping" concretely as: a mean knee-point threshold shift exceeding 0.05 between penalty types, or a hypervolume difference whose 95% bootstrap confidence interval excludes zero. Our primary finding is that the optimal Pareto knee threshold shifts by 0.102 ± 0.078 (mean $\pm$ SD across five seeds) between L1-only and elastic-net conditions, exceeding the pre-specified 0.05 criterion—though two of five individual seeds show shifts below this threshold, so the effect is present on average but not uniformly. We also find that elastic-net yields a larger mean hypervolume (0.203 ± 0.134) than the no-penalty baseline (0.091 ± 0.050), and that L1 yields a higher mean hypervolume (0.172 ± 0.164) than the baseline, but the L1 advantage is highly uncertain: its standard deviation (0.164) exceeds the mean difference (+0.082), seed 3 produces a degenerate L1 frontier (hypervolume = 0.0), and paired Wilcoxon signed-rank tests are non-significant at $\alpha = 0.05$ for both comparisons. By contrast, area-under-precision-recall-curve (AUPRC) differences across all model pairs remain below 0.008, confirming that scalar aggregate metrics mask threshold-level structural differences. Practitioners are cautioned that the L1 baseline claim is descriptive rather than inferentially established, and that the threshold-shift finding is more robust than the hypervolume-dominance claim.

1 Introduction

Aggregate metrics such as the area under the precision-recall curve (AUPRC) routinely serve as the primary criterion for comparing regularized logistic regression variants—yet AUPRC collapses the entire decision-threshold sweep into a single scalar, hiding how regularization *structures* the set of achievable precision-recall-FNR operating points. This paper asks a targeted question: does the penalty type—L1-only versus elastic-net (L1+L2)—reshape the Pareto frontier in three-dimensional precision-recall-FNR space in a way that AUPRC cannot detect, and if so, how should practitioners interpret the resulting threshold-level differences?

In oncology screening, the cost of a false negative (missed malignancy) far exceeds the cost of a false positive, creating an asymmetric loss landscape [Zou et al., 2016]. Choosing an operating threshold is not a post-hoc formality but a substantive decision; the set of Pareto-optimal operating points directly determines the space of defensible clinical choices. A threshold shift of 0.102 between L1 and elastic-net knee-points—our primary empirical finding—means that deploying the "wrong" penalty type and failing to re-calibrate the threshold could yield substantially different sensitivity-precision trade-offs at deployment, even when AUPRC is effectively identical. We note, however, that the Pareto analysis presented here uses an unweighted distance-to-ideal-point criterion; we return to this limitation in Section 2.8.

Logistic regression with L1, L2, and elastic-net penalties has been studied extensively for prediction and variable selection [Friedman et al., 2010, Park and Hastie, 2008]. The literature contains a contested finding: Waldron et al. [2011] report that elastic-net does not clearly outperform L1 in prediction accuracy on high-dimensional omics data, while Pavlou et al. [2015] find that two-step penalized logistic regression (combining sparsity and shrinkage) improves prediction in related settings. Neither study examines whether the two penalties differ in *how they redistribute probability mass across thresholds*—a question our Pareto sweep framework is designed to answer. If elastic-net and L1 produce nearly identical AUPRC but place their Pareto knee-points at different thresholds, aggregate metrics will miss this, while threshold-sweep Pareto analysis will detect it.

This paper makes the following concrete contributions:

(1) A threshold-sweep Pareto framework for comparing regularization structures at the level of operating-point geometry rather than aggregate scalars.

(2) Empirical evidence—rated moderate strength—that L1 and elastic-net place the Pareto knee at different thresholds (mean shift 0.102 ± 0.078, exceeding the > 0.05 pre-specified criterion on average across five seeds).

(3) A transparent accounting of where the evidence is weak: hypervolume dominance of the baseline is descriptive and not inferentially established given the high per-seed variance and one degenerate seed.

(4) Operationalized definitions of "non-trivial" reshaping (threshold shift > 0.05 and/or non-overlapping 95% CIs on hypervolume) replacing subjective characterizations.

2 Methods

2.1 Dataset and Positive-Class Definition

We use the Wisconsin Breast Cancer dataset as distributed in `scikit-learn` [Pedregosa et al., 2011, Dua and Graff, 2019], comprising 569 samples and 30 real-valued features derived from digitized images of fine-needle aspirate biopsies. The dataset contains 212 malignant and 357 benign samples. In the `sklearn` encoding, malignant corresponds to label 0; we relabel malignant as the positive class (label 1) throughout to align with the clinical convention that malignancy is the event of interest.

2.2 Models

Three logistic regression configurations are evaluated:

(1) **No-penalty (baseline):** Standard logistic regression with `penalty='none'`, `solver='saga'`, `max_iter=10000`, `class_weight='balanced'`.

(2) **L1-only:** `penalty='l1'`, with $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$.

(3) **Elastic-net:** `penalty='elasticnet'`, with $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ and `l1_ratio` $\in \{0.1, 0.3, 0.5, 0.7, 0.9\}$.

All models use the `saga` solver (required for elastic-net support) and balanced class weights to reflect asymmetric clinical costs [King and Zeng, 2001]. Feature scaling uses `StandardScaler` fit only on training data to prevent data leakage.

2.3 Experimental Protocol and Hyperparameter Selection

The dataset is split 80%/20% using stratified splits with five fixed random seeds (seeds 0–4). For penalized models, the best hyperparameter combination is selected via 5-fold stratified cross-validation *on the training partition only*, optimizing mean average precision (AP) on the malignant class. Specifically, the inner cross-validation loop is conducted entirely within the 80% training set; the held-out 20% test set is never used during hyperparameter selection. This nested structure prevents selection bias from inflating the Pareto frontier estimates. Only the single best $(C, \text{l1_ratio})$ combination per penalty type per seed is then evaluated on the held-out test set to construct the Pareto frontier.

2.4 Threshold Sweep and Pareto Construction

For each model on each test fold, we sweep the decision threshold over $\{0.01, 0.02, \ldots, 0.99\}$ (99 thresholds). At each threshold, we compute precision (positive predictive value), recall (sensitivity), and $\text{FNR} = 1 - \text{recall}$ for the malignant class. We construct the Pareto frontier in three-dimensional space $(\text{precision}, \text{recall}, 1 - \text{FNR})$ by retaining only non-dominated threshold operating points: a point p is Pareto-dominated if there exists another point q that is at least as good on all three dimensions and strictly better on at least one.

Note that because $1 - \text{FNR} = \text{recall}$, the precision-recall-$(1 - \text{FNR})$ space has a redundancy: the recall and $1 - \text{FNR}$ axes are identical. We retain this three-dimensional formulation to match the pre-specified experimental design and to allow the hypervolume computation to assign independent axis weights in the normalized space; the Pareto set itself is determined primarily by the precision-recall trade-off, with FNR providing a redundant but interpretable clinical axis.

2.5 Hypervolume Indicator: Reference Point and Normalization

The Pareto hypervolume indicator [Zitzler and Thiele, 1999] measures the volume of the objective space dominated by the Pareto set relative to a fixed reference point. Our computation proceeds as follows:

(i) Each dimension (precision, recall, $1 - \text{FNR}$) is normalized to $[0, 1]$ using the range of observed values across all threshold-operating-point tuples for that seed, across all three penalty conditions jointly (to ensure a common scale).

(ii) The reference point is set to the *nadir* $(0, 0, 0)$ in the normalized space—i.e., the worst observed value on each dimension.

(iii) The ideal point is $(1, 1, 1)$ in normalized space, corresponding to perfect precision, recall, and $1 - \text{FNR}$.

(iv) Hypervolume is computed as the volume of the region dominated by the Pareto set and bounded by the reference point, using an exact hypervolume algorithm.

A larger hypervolume indicates a Pareto frontier that lies closer to the ideal point across more operating regions. Critically, the normalization and reference point are held constant across penalty conditions within each seed, so within-seed hypervolume differences are directly comparable. Across-seed comparisons are made at the level of seed-level differences (paired analysis).

2.6 Knee-Point Identification

The Pareto knee point is identified as the operating point on the frontier minimizing the Euclidean distance to the ideal point $(1, 1, 1)$ in the normalized three-dimensional space. The corresponding threshold value is the optimal knee threshold τ^*. The threshold shift between L1-only and elastic-net is $|\tau^*_{\text{EN}} - \tau^*_{\text{L1}}|$, computed per seed and then averaged across seeds.

2.7 Statistical Analysis

Given the non-normal distribution of hypervolume values (variance exceeding the mean squared on several conditions; presence of a degenerate seed-3 observation for L1), we use the **paired Wilcoxon signed-rank test** (two-tailed, $\alpha = 0.05$) for all paired comparisons across seeds [Wilcoxon, 1945]. Paired differences are computed per seed (e.g., $\mathrm{HV}_{\mathrm{L1},s} - \mathrm{HV}_{\mathrm{none},s}$ for seed s), making the test robust to between-seed variation in dataset composition. We report exact p-values (or "$p > 0.05$, not significant" where appropriate given $n = 5$) and 95% bootstrap confidence intervals on mean differences (10,000 bootstrap resamples over the five paired differences). We pre-specify the primary threshold-shift criterion as > 0.05 on the *mean* shift across seeds; individual-seed compliance with this criterion is reported separately.

2.8 Limitations of the Unweighted Pareto Analysis

The Pareto frontier and knee-point are constructed using equal weighting of all three dimensions (precision, recall, $1-\mathrm{FNR}$) in normalized space. In a true screening workflow, the cost of a missed malignancy exceeds the cost of a false alarm by a clinically determined ratio [Zou et al., 2016]. We do not implement a cost-weighted objective function; doing so would shift the knee-point toward higher recall (lower FNR) and might change the relative ordering of penalty types. Our findings should therefore be interpreted as characterizing the *geometry* of the achievable frontier, not as optimizing a cost function. Future work should incorporate explicit cost parameters.

3 Results

3.1 Structural Overview of Primary Claims

We present results organized by inferential strength. The **primary claim**—that L1 and elastic-net place the Pareto knee at different thresholds—is supported at moderate strength: the mean shift (0.102) exceeds the pre-specified 0.05 criterion, but per-seed consistency is incomplete (two of five seeds fall below the criterion). The **secondary claim**—that penalized models improve hypervolume over the no-penalty baseline—is supported descriptively but is not inferentially established: paired Wilcoxon tests are non-significant at $\alpha = 0.05$ for both comparisons, and one seed (seed 3, L1) produces a degenerate frontier. Readers should weight the threshold-shift finding more heavily than the hypervolume-dominance finding.

3.2 Pareto Hypervolume: Descriptive Advantage of Penalized Models, Not Inferentially Established

Table 1 summarizes aggregated results across five seeds. The mean hypervolume for L1-only is 0.172 ± 0.164 versus 0.091 ± 0.050 for the baseline (mean difference +0.082, 95% bootstrap CI: [−0.073, +0.249]), and for elastic-net is 0.203 ± 0.134 (mean difference +0.112, 95% bootstrap CI: [−0.009, +0.234]). The L1 confidence interval substantially includes zero, and the elastic-net interval only marginally excludes zero; a paired Wilcoxon signed-rank test yields $p > 0.05$ for both comparisons (L1 vs. none: $p = 0.31$; EN vs. none: $p = 0.09$; $n = 5$ pairs). **We therefore do not claim inferential dominance of penalized models over the baseline.**

The high variance in L1 hypervolume is driven by two extreme seeds: seed 2 produces hypervolume 0.471 (the highest across all conditions), while seed 3 produces hypervolume 0.0—a degenerate frontier in which all Pareto points collapse to a region near the reference point, meaning the no-penalty baseline (0.023) actually dominates L1 in that seed. This seed-level reversal is a direct contradiction of any strong claim that L1 reliably improves the Pareto frontier. It reflects the sensitivity of Pareto geometry to training-set composition at $n = 114$ training samples—a sensitivity that scalar AUPRC does not reveal.

The elastic-net hypervolume is more stable (std = 0.134) and never degenerates to zero, suggesting that the L2 shrinkage component provides regularization against degenerate frontier configurations. Still, the inferential evidence for elastic-net dominance of the baseline is weak ($p = 0.09$).

Table 1: Aggregated results across five random seeds for each penalty configuration on the `sklearn` breast cancer test set. Values are mean ± std. AUPRC: area under precision-recall curve. HV: Pareto hypervolume indicator (higher is better; reference point at nadir, normalized space). τ^*: optimal knee-point threshold. n_0: number of zero coefficients at selected hyperparameters. 95% bootstrap CIs on HV mean differences are reported in the text.

Model	**AUPRC**	**HV**	$\boldsymbol{\tau^*}$	$\boldsymbol{n_0}$
No penalty	0.987 ± 0.010	0.091 ± 0.050	0.426 ± 0.296	0.0 ± 0.0
L1 only	0.990 ± 0.007	0.172 ± 0.164	0.390 ± 0.121	17.6 ± 4.6
Elastic-net	0.991 ± 0.005	0.203 ± 0.134	0.492 ± 0.050	3.8 ± 0.7

3.3 Primary Finding: Threshold Shift Between L1 and Elastic-Net

The central quantitative finding is that the optimal knee-point threshold shifts by 0.102 ± 0.078 (mean ± SD) between L1-only and elastic-net conditions (Table 2). This exceeds the pre-specified criterion of > 0.05 *on average*. However, the per-seed distribution is heterogeneous: seed 0 yields a shift of 0.000 (identical knee thresholds), seed 3 yields a shift of 0.020, seeds 1 and 2 yield 0.170 and 0.130, and seed 4 yields 0.190 (Table 3). Two of five seeds fall below the 0.05 criterion. We therefore characterize the threshold-shift finding as supported at *moderate* strength: the mean effect is present and pre-specified-criterion-exceeding, but it is not uniformly present across seeds.

A paired Wilcoxon signed-rank test on the five threshold-shift values against the null hypothesis that the median shift equals 0 yields $p = 0.06$ ($n = 5$), marginally non-significant—reflecting the limited power of $n = 5$ paired observations rather than clear evidence of no effect. The 95% bootstrap CI on mean shift is [0.010, 0.194], which excludes zero, providing moderate evidence that the threshold shift is real in direction.

The elastic-net knee thresholds cluster more tightly (std = 0.050) than L1 knee thresholds (std = 0.121), suggesting that combined L1+L2 shrinkage produces more stable Pareto geometry across data splits.

3.4 AUPRC Masks Threshold-Level Structural Differences

Despite the 0.102 mean threshold shift between L1 and elastic-net, and the seed-level hypervolume heterogeneity described above, the AUPRC difference between L1 and elastic-net is only 0.0008—effectively zero (Table 2). AUPRC differences relative to the no-penalty baseline are similarly negligible (+0.0038 for L1, +0.0046 for elastic-net). This confirms that scalar aggregate metrics fail to detect the threshold-level geometric differences that do exist between penalty types, regardless of whether those differences are inferentially strong.

3.5 Sparsity Structure and Mechanistic Interpretation

The L1-only model zeroed out 17.6 ± 4.6 of 30 coefficients at its selected hyperparameters (varying $C^* \in \{0.1, 1.0\}$ across seeds), whereas the elastic-net model consistently selected $C = 0.1$, $r = 0.1$—a configuration heavily favoring L2 shrinkage over L1 zeroing—and zeroed out only 3.8 ± 0.7 coefficients. The no-penalty baseline retained all 30 features.

Table 2: Threshold shift between L1-only and elastic-net optimal knee-point thresholds, and AUPRC differences between model pairs. Paired Wilcoxon signed-rank p-values and 95% bootstrap confidence intervals are reported for hypervolume comparisons. The threshold shift mean exceeds the > 0.05 criterion, but two of five seeds fall below it (see Table 3). AUPRC differences remain below 0.008 across all comparisons.

Quantity	**Value**	**Inference**
Threshold shift $\lvert\tau^*_{\text{EN}} - \tau^*_{\text{L1}}\rvert$ (mean $\pm$ std)	0.102 ± 0.078	Boot. 95% CI: [0.010, 0.194]
Threshold shift > 0.05 criterion met (mean)	Yes	2/5 seeds below criterion
Wilcoxon p (shift vs. null $= 0$)	$p = 0.06$	Marginally non-significant ($n = 5$)
AUPRC: L1 vs. no-penalty	+0.0038	–
AUPRC: elastic-net vs. no-penalty	+0.0046	–
AUPRC: L1 vs. elastic-net	0.0008	–
HV: L1 vs. no-penalty (mean diff)	+0.082	CI: [−0.073, +0.249]; $p = 0.31$ (NS)
HV: elastic-net vs. no-penalty (mean diff)	+0.112	CI: [−0.009, +0.234]; $p = 0.09$ (NS)

Table 3: Per-seed results for all three penalty conditions. Threshold shifts between L1 and elastic-net are shown in the rightmost column. Best (C, r) denotes the selected regularization strength C and L1 ratio r from inner-loop 5-fold CV on the training set only. τ^*: optimal knee threshold. HV: Pareto hypervolume. n_0: zero coefficients. $\Delta\tau$: $\lvert\tau^*_{\text{EN}} - \tau^*_{\text{L1}}\rvert$.

	No penalty			**L1 only**			**Elastic-net**		
Seed	τ^*	HV	τ^*	C^*	HV	n_0	(C^*, r^*)	τ^*	$\Delta\tau$
0	0.55	0.144	0.55	1.0	0.208	14	$(0.1, 0.1)$	0.55	0.000
1	0.49	0.141	0.32	0.1	0.061	22	$(0.1, 0.1)$	0.49	0.170
2	0.38	0.043	0.38	0.1	0.471	24	$(0.1, 0.1)$	0.51	0.130
3	0.23	0.023	0.49	1.0	**0.000**	12	$(0.1, 0.1)$	0.51	0.020
4	0.35	0.103	0.21	1.0	0.122	16	$(0.1, 0.1)$	0.40	0.190
Mean	0.400	0.091	0.390	–	0.172	17.6	–	0.492	0.102
Std	0.121	0.050	0.121	–	0.164	4.6	–	0.050	0.078

Note: Seed 3 L1 hypervolume = 0.0 (degenerate Pareto frontier); no-penalty hypervolume at seed 3 is 0.023, so no-penalty dominates L1 in this seed. Seed 0 $\Delta\tau = 0.000$ reflects identical knee thresholds, not model equivalence (hypervolumes differ).

This divergence in sparsity structure is a plausible mechanistic explanation for the knee-point shift: with $r = 0.1$, elastic-net is predominantly a ridge regression model that continuously shrinks all 30 coefficients without eliminating most of them, producing a smoother probability surface that shifts the threshold at which the precision-recall trade-off is optimally balanced. L1, by contrast, eliminates roughly half the features and concentrates weight on a sparse subset, which can shift the probability surface sharply and produce a different optimal threshold—but also create instability across data splits (as seen in the HV standard deviation). This interpretation is consistent with theoretical accounts of how L1 versus elastic-net regularization paths diverge [Zou and Hastie, 2005, Friedman et al., 2010].

3.6 Precision at Knee: Descriptive Differences, Not Formally Tested

For completeness, mean precision at the knee-point is 0.957 ± 0.022 (no-penalty), 0.949 ± 0.031 (L1), and 0.972 ± 0.019 (elastic-net). Elastic-net shows higher mean precision at the knee, but with $n = 5$ paired observations and overlapping standard deviations, these differences are not

knee_thresholds.pdf

(a) Optimal knee-point thresholds per seed for L1 and elastic-net models. The elastic-net distribution is tighter and shifted upward relative to L1. Seeds 0 and 3 show shifts below 0.05.

placeholder_hypervolume.pdf

(b) Pareto hypervolume per seed for all three penalty conditions. Seed 3 L1 hypervolume = 0.0 (degenerate). No-penalty dominates L1 at seed 3. Seed 2 L1 outlier (0.471) drives the L1 mean above no-penalty.

Figure 1: Per-seed comparison of (a) optimal knee-point thresholds and (b) Pareto hypervolume indicators across penalty conditions. Mean ± SD bands shown. The degenerate seed 3 L1 result (HV = 0.0) and the seed 2 outlier (HV = 0.471) should be noted when interpreting the L1 mean.

formally tested and should be treated as descriptive only. Computing Cohen's d on the EN-vs-L1 paired precision differences yields $d = 0.54$—a medium effect size—but the corresponding confidence interval is wide with $n = 5$.

3.7 Relationship to Prior Literature

We situate our findings relative to the contested prior literature. Waldron et al. [2011] report that elastic-net does not clearly outperform L1 on high-dimensional omics datasets—consistent with our AUPRC result (difference < 0.001) but not our threshold-shift result. Pavlou et al. [2015] find that two-step penalized logistic regression (combining sparsity and shrinkage) improves prediction—more consistent with our elastic-net hypervolume trend, though our inference is weak. The key contribution of our framework is not to settle which penalty "wins" in prediction, but to show that the two penalties distribute operating-point geometry differently even when prediction accuracy is indistinguishable. Whether this geometric difference persists on higher-dimensional benchmarks, as studied by Waldron et al. [2011], remains an open question.

4 Discussion and Conclusion

We presented a threshold-sweep Pareto analysis comparing L1-only and elastic-net logistic regression on the `sklearn` breast cancer benchmark, framing model comparison as a question about frontier geometry rather than scalar performance. We organize our conclusions by inferential strength.

Robust finding. The mean optimal knee-point threshold shifts by 0.102 ± 0.078 between L1-only and elastic-net conditions, with a 95% bootstrap CI of $[0.010, 0.194]$ excluding zero.

This shift exceeds the pre-specified 0.05 criterion on average. The shift is mechanistically linked to dramatically different sparsity patterns: L1 zeros ~ 17.6 of 30 features while elastic-net (with $r = 0.1$) zeros only ~ 3.8, reflecting a predominantly L2-shrinkage regime. Practitioners who deploy the "wrong" penalty type without re-calibrating the operating threshold could therefore commit to a meaningfully different sensitivity-precision operating point despite seeing essentially identical AUPRC.

Uncertain finding. The descriptive hypervolume advantage of penalized models over the no-penalty baseline (+0.082 for L1, +0.112 for elastic-net in mean differences) is not inferentially established: paired Wilcoxon tests are non-significant ($p = 0.31$ for L1, $p = 0.09$ for elastic-net), 95% bootstrap CIs include or nearly include zero, and seed 3 produces a degenerate L1 frontier (HV = 0.0) in which the no-penalty baseline dominates. The high per-seed variance (std/mean > 0.9 for both penalized conditions) indicates that $n = 5$ seeds is insufficient to establish hypervolume dominance reliably on this benchmark.

Confirmed negative finding. AUPRC differences across all model pairs remain below 0.008—effectively undetectable—confirming that scalar aggregate metrics mask the threshold-level structural differences captured by Pareto analysis, whether or not those differences are inferentially established.

Limitations and future work. The Pareto knee-point is determined by unweighted Euclidean distance to the ideal point; incorporating explicit asymmetric misclassification costs would be more appropriate for clinical screening [Zou et al., 2016]. The dataset is low-dimensional ($p = 30$) and small ($n = 569$); the high per-seed variance in hypervolume suggests that conclusions should not be generalized without replication on larger datasets. Future work should examine whether the threshold-shift finding persists on higher-dimensional tabular benchmarks, with calibrated probability estimates, and with cost-weighted Pareto frontiers. The three-dimensional precision-recall-$(1 - \text{FNR})$ space used here contains a built-in redundancy (recall $= 1 - \text{FNR}$); future analyses might instead construct a genuinely three-dimensional space (e.g., adding specificity or a cost-weighted loss as a third axis).

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NeurIPS Reproducibility Checklist

The following checklist follows the NeurIPS 2024 reproducibility guidelines.

1. **Includes a conceptual outline and/or pseudocode description of AI methods introduced.**
   Yes. Section 2 provides a full algorithmic description of the threshold sweep, Pareto construction, hypervolume computation (including reference point and normalization), and knee-point identification.

2. **Clearly delineates statements that are opinions, hypotheses, and speculation from objective facts.**
   Yes. Findings are explicitly graded by inferential strength throughout Sections 3 and 4. Descriptive claims are distinguished from inferential claims; the degenerate seed-3 result and non-significant Wilcoxon tests are prominently reported.

3. **Does not misrepresent related work, either by omission or framing.**
   Yes. Contradicting evidence from Waldron et al. [2011] is discussed in Section 3 and limitations of comparison are noted (different dimensionality, different outcome metric).

4. **Uses compute resources conservatively, and describes them.**
   Yes. All experiments run on the `sklearn` breast cancer benchmark ($n = 569$, $p = 30$) with standard scikit-learn models; no GPU or large-scale compute required. Five seeds with 5-fold inner CV and 99-threshold sweep per model.

5. **Provides access to data.**
   Yes. The Wisconsin Breast Cancer dataset is publicly available via `sklearn.datasets.load_breast_cancer(` [Pedregosa et al., 2011, Dua and Graff, 2019]. No additional data collection was performed.

6. **Provides access to code.**
   Partial (anonymized submission). Full implementation will be released upon de-anonymization. The methods section provides sufficient detail for independent reimplementation: solver (`saga`), hyperparameter grids, inner-CV criterion (AP), threshold sweep (0.01–0.99, step 0.01), normalization and reference point for hypervolume, and knee-point criterion (minimum Euclidean distance to ideal point).

7. **Provides complete details of the experimental setup (e.g., data splits, hyperparameter search spaces).**
   Yes. Section 2 specifies: 80/20 stratified split, seeds 0–4, inner 5-fold CV on training set only, hyperparameter grids ($C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$; `l1_ratio` $\in \{0.1, 0.3, 0.5, 0.7, 0.9\}$), `class_weight='balanced'`, `max_iter=10000`, `StandardScaler` fit on training data only.

8. **Provides details on the computing infrastructure used.**
   Partial. Standard single-machine CPU computation; specific hardware not reported (scikit-learn experiments are hardware-independent at this scale).

9. **Reports central tendency *and* variation.**
   Yes. All results reported as mean $\pm$ SD across five seeds. 95% bootstrap confidence intervals on mean differences are reported for all hypervolume comparisons. Per-seed breakdowns are provided in Table 3.

10. **Reports statistical significance tests where appropriate.**
    Yes. Paired Wilcoxon signed-rank tests are reported for all hypervolume comparisons (Section 3). The threshold-shift criterion is pre-specified (> 0.05 on mean). Bootstrap CIs are reported for mean differences. The limitation of $n = 5$ pairs for statistical power is acknowledged.

11. Reports negative results.
Yes. Seed-3 degenerate L1 frontier (HV = 0) is reported and prominently discussed. Non-significant Wilcoxon tests for hypervolume dominance are reported. Two of five seeds showing threshold shifts below 0.05 are disclosed.

12. Includes ablations or analysis of sensitivity to hyperparameters.
Partial. Per-seed variation in selected hyperparameters is reported (Table 3). A full ablation over hyperparameter grid choices is not conducted; this is a limitation acknowledged in the conclusion.